```
* NOTE: idt() = integration function
.SUBCKT Block1_fixe_Drehzahl U1 V1 W1 F1 M n GND1
+  PARAMS:
+   UN_verkettet=20000
+   IN=14.434k
+   Pel=400e6  ;UN_verkettet*IN*sqrt(3)*cosphi of 0.8
+   fn=50
+   Kc=0.5
+   UE0=600V ; Power of exciter = 0.25-5% * SN machine (smaller values for higher SN)
+   IE0=2000A
+   p=2
+   Streuung=0.05 ;
+   rcu=0.001
+   rbg=0.001
+  Schenkelpolfaktor=1 ;(1=cylindrical pole 0.5=Salient pole)
+  Daempferfaktor=1
+  err0=0
+  RE_factor=0.6666
+  n0=1500
+  n1={n0} ;starts with 1500U/min
+  alpha_0=0
+  a=0.9 ; Parameter for nonlinear noload line (should be about 0.6-0.9)
+  m=7
.PARAM
+  a1={a};  Saturation parameter d-axis
+  m1={m};  Saturation parameter d-axis
+  a2={a1}; Saturation parameter q-axis
+  m2={m1}; Saturation parameter q-axis
+  UN={UN_verkettet/sqrt(3)}
+  ZN={UN/IN}
+  Rs={ZN*rcu}
+  omega_el={(2*pi)*fN}
+  Ls={Streuung*ZN/Kc/omega_el}
+  PsiN={sqrt(2)*UN/omega_el}
+  RE={UE0/IE0}
+  RD={Rs/Daempferfaktor}
+  RDx={RD} ; can be overwritten by setting this parameters on modelline
+  RDy={RD} ; can be overwritten by setting this parameters on modelline
+  LDx={Ls} ; can be overwritten by setting this parameters on modelline
+  LDy={Ls} ; can be overwritten by setting this parameters on modelline
+  LE={Ls} ; can be overwritten by setting this parameters on modelline
+  R0={Rs} ; can be overwritten by setting this parameters on modelline
+  L0={Ls} ; can be overwritten by setting this parameters on modelline (X0 is 0.03-0.1 p.u)
+  Lsd={Ls}
+  Lsq={Ls}
+  XD={UN/IN/Kc}
+  XQ={XD*Schenkelpolfaktor}
+  g={UE0/(UN*sqrt(2)*RE/(XD-omega_el*Lsd))} ; updated formula for g parameter
+  g1={UE0/(UN*sqrt(2)*RE/(XQ-omega_el*Lsq))} ; updated formula for salient pole machine
+  ImNx={IE0/g} ; d-axis magnetizing current
+  ImNy={IE0/g1} ;q-axis magnetizing current
+  MN={Pel*(1-rbg-rcu)*p/omega_el}
*
*           currents of windings
G1    U1   U2     VALUE={v(isxS)+v(i_Null)}
G2    V1   V2     VALUE={(-v(isxS)+v(isyS)*sqrt(3))/2+v(i_Null)}
G3    W1   W2     VALUE={(-v(isxS)-v(isyS)*sqrt(3))/2+v(i_Null)}
```

```
GiE          F1          0          VALUE={v(ie)*g}
*            voltage components
EuxS   usxS   0   VALUE={(2*v(u1,u2)-v(v1,v2)-v(w1,w2))/3}
EuyS   usyS   0   VALUE={(v(v1,v2)-v(w1,w2))/sqrt(3)}
Eux    usx    0   VALUE={ v(usxS)*cos(v(alpha))+v(usyS)*sin(v(alpha))}
Euy    usy    0   VALUE={-v(usxS)*sin(v(alpha))+v(usyS)*cos(v(alpha))}
* Starpoint
R2    v2    GND1   1u
R12   u2    GND1   1u
R23   w2    GND1   1u
RGND1  GND1  0  1e6
*            Zerosystem
Eu0  u_Null 0  VALUE={(v(u1,u2)+v(v1,v2)+v(w1,w2))/3}
Ei0   i_Null  0  VALUE={idt(v(u_Null)-R0*v(i_null))/L0}
*            current components
EisxS  isxS  0 VALUE={v(isx)*cos(v(alpha))-v(isy)*sin(v(alpha))}
EisyS  isyS  0 VALUE={v(isy)*cos(v(alpha))+v(isx)*sin(v(alpha))}
*            magnetic coupling of direct axis with saturation
Gpsix  0  isx VALUE={v(psisx)}
Rsx    isx  0  {1/Lsd}
Gpsidx isx iDx1 VALUE={v(psiDx)}  ;
Rsdx   iDx1 isx  {1/LDx}
GpsiEx iDx1 imx1 VALUE={v(psiE)}
RsEx   imx1 iDx1 {1/LE}
vphx    imx1 imxa1 0
Eimx   imxa1  0  VALUE={ImNx*(a1*i(vphx)/PsiN+(1-a1)*PWRS(i(vphx)/PsiN,m1))}
*            magnetic coupling of quadrature axis with saturation
Gpsiy   0  isy  VALUE={v(psisy)}
Rsy     isy   0  {1/Lsq}
Gpsidy  isy   imy1  VALUE={v(psiDy)}
Rsdy    imy1  isy  {1/LDy}
vphy    imy1   imya1  0
Eimy    imya1  0  VALUE={ImNy*(a2*i(vphy)/PsiN+(1-a2)*PWRS(i(vphy)/PsiN,m2))}
*            damper currents
Eidx  idx 0 VALUE={v(idx1)-v(isx)}
Eidy  idy 0 VALUE={v(imy1)-v(isy)}
*            exciter current
Eie ie 0 VALUE={v(imx1)-v(idx1)}
*            Flux components
Epsi_sx  psisx   0 VALUE={err0*PsiN+idt(v(usx)-Rs*v(isx)+p*(2*pi)/60*v(n)*v(psisy))}
Epsi_sy  psisy   0 VALUE={idt(v(usy)-Rs*v(isy)-p*(2*pi)/60*v(n)*v(psisx))}
Epsi_Dx  psiDx  0 VALUE={err0*psiN+idt(-RDx*v(idx))}
Epsi_Dy  psiDy  0 VALUE={idt(-RDy*v(idy))}
Epsi_E   psiE  0 VALUE={err0*psiN*(1+Streuung)+idt(v(F1,0)*g*RE_factor-RE*RE_factor*g*g*v(ie))}
*            Load angle
Etheta 0 theta VALUE={idt(180/pi*atan2(v(usx),v(usy)))}
*            inner torque
Emi  Mi  0 VALUE={1.5*p*(i(vphx)*v(isy)-i(vphy)*v(isx))}
*            outher torque
EMoment  M 0 VALUE={-v(Mi)+v(Rbg)}
Vn n 0 {n1}
* friction
GRbg   0  Rbg  TABLE {v(n)} (-1,-1) (1,1)
RRbg   Rbg   0  {MN*rbg}
* Compute angle alpha
Ealpha alpha  0  VALUE={alpha_0*pi/180+idt(v(n))*p/60*(2*pi)}
.ENDS
```